Probability & Statistics: colloquium II
Random variables

1 Basic definitions and notions

1. Discrete random variable
2. Probability mass function (p.m.f.)
3. Cumulative distribution function (c.d.f.)
4. Binomial distribution
5. Geometric distribution
6. Poisson distribution
7. Continuous random variable
8. Probability density function (p.d.f.)
9. Uniform distribution
10. Exponential distribution
11. Normal distribution
12. Expectation and moments
13. Variance and covariance
14. Correlation and its properties
15. Random vectors and their joint distributions
16. Marginal distributions
17. Conditional distributions
18. Distribution of a transformed random variable
19. Distribution of the sum of r.v.
20. Probability generating function (p.g.f.)
21. Moment generating function (m.g.f.)

2 List of questions

1. Discrete random variables; probability mass function and cumulative distribution function and their properties; examples
2. Binomial distribution; Geometric distribution; Poisson distribution; examples
3. Joint and marginal distributions; conditional distribution; examples
4. Expectation of a random variables and its properties; examples
5. Moments and variance; properties of variance
6. Covariance and correlation
7. Probability generating functions and their properties
8. Continuous random variables; their properties
9. Independent random variables; examples and properties
10. Normal distribution and its properties
11. Exponential distribution and its properties
12. Independent random variables

3 Seminar 4

Problem 1 (Discrete r.v. and their p.m.f.)**.** Five men and five women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman and let Y denote their lowest ranking. (For instance, $X = 1$ if the top-ranked person is a female.)

(a) Find the probability mass function (p.m.f.) of a random variable X;

(b) explain how to find the joint p.m.f. of a random vector (X, Y), i.e., how to calculate the values $p_{(X,Y)}(x, y) := \mathsf{P}(X = x, Y = y)$ for all pairs of x and y;

(c) what is the marginal distribution of X? Calculate the value of the marginal distribution of X for $x = 3$ and compare to the value found in part (a);

(d) are the random variables X and Y independent?

Problem 2 (Discrete r.v. and their p.m.f.)**.** Suppose that a fair die is rolled twice. What is the probability mass function of the following random variables:

(a) $X =$ ‘‘`the maximum value to appear in the two rolls`’’;

(b) $Y =$ ‘‘`the minimum value to appear in the two rolls`’’. Give a short explanation why $p_X(j) = p_Y(7 - j)$ for $j = 1, 2, \ldots, 6$.

(c) Find the joint p.m.f of the random vector (X, Y); calculate the marginal distribution of X and compare to part (a).

(d) Are random variables X and Y independent?

Problem 3 (Binomial distribution)**.** A communications channel transmits the digits 0 and 1. However, due to noise, the digit transmitted is incorrectly received with probability 0.2. Suppose that we want to transmit an important message consisting of n binary digits. To reduce the chance of error, we transmit 00000 instead of 0 and 11111 instead of 1. If the receiver of the message uses the "majority" rule for decoding, what is the probability that

(a) the message will be wrong when decoded?

(b) at most two binary digits of the original message will be wrongly decoded?

(c) what is the most probable number of correctly decoded binary digits?

What independence assumptions are you making?

Problem 4 (Binomial distribution)**.** When coin A is flipped, it lands on Heads with probability 0.4; when coin B is flipped, it lands on Heads with probability 0.6. One of these coins is randomly chosen and flipped 10 times; let X denote the total number of Heads.

(a) Is it true that a r.v. X has a binomial distribution?

(b) Find the p.m.f. of the random variable X.

(c) Given that the first of these ten flips lands Heads, what is the conditional distribution of the total number of Heads? (Hint: condition on the coin chosen)

(d) Assuming that $X = k$, what is the (conditional) probability that coin A is tossed? For what k is it larger than 0.5?

Problem 5 (Binomial distribution)**.** Suppose that a biased coin that lands on Heads with probability p is flipped 10 times. Given that a total of 6 Heads results, find the conditional probability that the first 3 outcomes

(a) are HTT (meaning that the first flip results in Heads, the second in Tails, and the third in Tails);

(b) include precisely k Heads; $k = 0, 1, 2, 3$.

Assume now that there are 2 Heads in the first three tosses of the coin.

(c) What is the conditional probability that there are k Heads in all 10 tosses?

(d) What is the most probable number of Heads in all 10 tosses?

Problem 6 (Poisson distribution)**.** Suppose that the number of new COVID-19 positive tested patients in the city A on day i is a Poisson random variable X_i of parameter $\lambda = 10$; assume also that the random variables $X_1, X_2, \dots$ are independent.

(a) Assume that by this noon, 5 new cases were reported. Find the probability that k newly COVID-19 infected patients will be reported today in total.

(b) Find the probability that the number of newly COVID-19 infected patients tomorrow will be larger than today's count.

Assume that the number of new COVID-19 positive tested patients in the city B on day i is a Poisson random variable Y_i of parameter $\mu = 20$; assume also that all random variables X_i and Y_j, $i, j = 1, 2, \dots$ are independent.

(c) Show that the random variable $X_i + Y_i$ has Poisson distribution of parameter 30.

(d) Assume that 30 new COVID-19 infected patients were reported yesterday in the cities A and B in total. What is the probability that k of them come from A, $k = 0, 1, \dots, 30$?

Problem 7 (Joint probability distribution)**.** A fair coin is tossed three times. Denote by X the number of Heads in the first two tosses and by Y the number of Tails in the last two tosses of the coin.

(a) Find the joint probability mass function $p_{X,Y}$ of X and Y:

Events	$X=0$	$X=1$	$X=2$
$Y=0$	...	...	...
$Y=1$	...	...	...
$Y=2$	...	...	...

prob.	$X=0$	$X=1$	$X=2$
$Y=0$	...	...	...
$Y=1$	...	...	...
$Y=2$	...	...	...

(b) Calculate the marginal pmf's of X and Y

(c) What distributions do X and Y have?

(d) Are X and Y independent?

(e) Calculate the conditional pmf of X given that $Y = 1$

Problem 8 (Joint probability distribution)**.** You roll two fair dice, and they show X and Y, respectively. Let $U = \min\{X, Y\}$, $V = \max\{X, Y\}$.

(a) Write down the joint distributions of:

(a) $\{U, X\}$; (b) $\{U, V\}$; (c) $\{X, Y, V\}$

(b) Which of the above **r.v.** are independent?

(c) Find the conditional **p.m.f.** of the **r.v.** $X \mid U = u$ and $Y \mid V = v$.

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Problem 9. If $\mathsf{E}(X) = 1$, $\mathsf{E}(Y) = -1$, $\mathsf{Var}(X) = 4$, and $\mathsf{Var}(Y) = 3$, find

(a) $\mathsf{E}\big((X-1)\ln 2\big)$; (b) $\mathsf{E}(3X^2-5X)$; (c) $\mathsf{E}((-4-Y)Y)$; (d) $\mathsf{Var}((\frac{1}{2}-X)X)$.

Problem 10. If $\mathsf{E}(X) = 2$, $\mathsf{E}(Y) = -1$, $\mathsf{E}(X^2) = 5$, and $\mathsf{E}(Y^2) = 4$, find

(a) $\mathsf{E}\big(X - eY + e^2\big)$; (b) $\mathsf{Var}(\pi + \sqrt{3}Y)$; (c) $\mathsf{E}((5-X)Y)$; (d) $\mathsf{Var}(\frac{1}{2}X + \frac{1}{3}Y)$.

Problem 11. An urn contains 10 balls numbered 1 to 10. Two balls are sampled in turn without replacement and their numbers are denoted as X_1 and X_2 respectively. Let Y be the minimum of X_1 and X_2.

(a) Show that the distributions of X_1 and X_2 coincide.

(b) Find expectation and variance of X_1.

(c) Find covariance of X_1 and X_2. Can you guess the sign before calculations? Are X_1 and X_2 independent?

(d) Find expectation and variance of Y.

(e) Find the conditional distribution of X_1 given that $Y = y$.
Hint: the case $X_1 = y$ is special

(f) Find expectation and variance of X_1 given that $Y = y$.

Problem 12. A fair coin is tossed repeatedly and independently until two consecutive heads or two consecutive tails appear. Denote by N the r.v. equal to the number of tosses made.

(a) Find the p.m.f. of the r.v. N.

(b) Find the expected value of the r.v. N.

(c) Find the variance of the r.v. N.

Assume now that the coin is tossed until we obtain a tail that is immediately preceded by a head; let a r.v. N_1 denote the corresponding number of tosses.

(d) Find the p.m.f. and the expected value of the r.v. N_1.

Problem 13. A die with 6 faces, numbered $1, \ldots, 6$, is rolled a fixed number of times n. The probability that the ith face comes up on each roll is denoted p_i and the results of different rolls are assumed independent. Let X_i be the number of times that the ith face comes up.

(a) Find the joint p.m.f. $p_{X_1,\ldots X_r}(k_1, \ldots, k_r) = \mathsf{P}(X_1 = k_1, \ldots, X_r = k_r)$ for $r = 1, 2$, and 6.

(b) Find the expected value of X_i.

(c) Find the variance of X_i.

(d) Find $E[X_i X_j]$ for $i \neq j$.

Problem 14. Assume that X is a **r.v.** with geometric distribution $\mathscr{G}(p)$.

(a) Calculate the *probability generating function* (p.g.f.) G_X of X.

(b) Use G_X to find $\mathsf{E}(X)$, the second factorial moment $\mu^{(2)}(X)$ and $\mathsf{Var}(X)$

Assume further that $X_1, X_2, \ldots$ are r. v.'s with geometric distribution $\mathscr{G}(p)$ and N is a r. v. with Poisson distribution $\mathscr{P}(\lambda)$. Assume that the r. v.'s $N, X_1, X_2, \ldots$, are independent and set $S := X_1 + \cdots + X_N$.

(c) Calculate the p.g.f. G_S of the r. v. S.
Hint: condition on the event $\{N = n\}$, $n \in \mathbb{Z}_+$

Problem 15. The number of people that enter an elevator at a certain time is a Poisson random variable with parameter $\lambda = 10$. The weight of each person (rounded to integer lbs) is independent of every other person's weight, and is uniformly distributed between 100 and 200 lb. Let X_i be the fraction of 100 by which the ith person exceeds 100 lb, e.g., if the 7th person weighs 175 lb, then $X_7 = 0.75$. Let Y be the sum of the X_i.

(a) Find the p.g.f. G_Y of the r. v. Y.

(b) Use G_Y to compute the expected value of Y.

(c) Verify your answer to part (b) by using the *total probability law* for expectations.

Problem 16. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win \$2 for each black ball selected and we lose \$1 for each white ball selected. Let X denote our winnings. Find the expected value and variance of the random variable X.

Problem 17. Suppose that a fair die is rolled twice. Find *expectation* and *variance* for the following random variables:

(a) the maximum value to appear in the two rolls;

(b) the minimum value to appear in the two rolls;

(c) the value of the first roll minus the value of the second roll.

Problem 18. Assume X and Y have joint probability mass function

$p(x,y)$	$y=-1$	$y=1$	$y=2$	$y=3$
$x=0$	1/12	1/4	0	0
$x=1$	1/12	1/12	1/12	1/12
$x=3$	0	0	1/4	1/12

(a) Find $\mathsf{E}(X)$, $\mathsf{E}(Y)$, $\mathsf{E}(S)$, $\mathsf{E}(D)$, and $\mathsf{E}(M)$, where $S = X + Y$, $D = X - Y$, and $M = \max\{X, Y\}$.

(b) Find whether X is independent of Y.

(c) Find variance of the random variables S and M.

(d) Find covariance of X and D.

(e) Find $\mathsf{E}(Y \mid X = x)$ for $x = 0, 1, 3$.

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Problem 19. Assume that Z is a standard normal **r.v.** Find the **p.d.f.** of the following **r.v.**'s:
(a) $2Z - 3$; (b) $|Z|$; (c) Z^2.

Problem 20. (2pts) Assume that X_1, X_2, ..., X_n are independent **r.v.**'s with distribution $\mathscr{U}(0, a)$. Find distribution (**c.d.f.** and **p.d.f.**) of the **r.v.**'s
(d) $U \sim \max\{X_1, \ldots, X_n\}$; (e) $V \sim \min\{X_1, \ldots, X_n\}$

Problem 21. Suppose that X and Y have joint probability density function

$$f_{X,Y}(x, y) = \begin{cases} 9e^{-3x-3y} & x \geq 0, y \geq 0; \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the marginal **p.d.f.**'s of X and Y; prove that X and Y are independent.

(b) find the joint distribution of the **r.v.**'s $S := X + Y$ and Y;

(c) find the distribution of S

(d) find the joint distribution of the **r.v.**'s $D := X - Y$ and Y;

(e) find the distribution of D.

Hint: in (b) and (d), use the formula for the **p.d.f.** of a transformed random vector $(X \pm Y, Y)$

Problem 22. Calculate the *moment generating function* (**m.g.f.**) $g_X(t) := \mathsf{E}(e^{tX})$ of a **r.v.** X with the

(a) uniform distribution $\mathscr{U}(a, b)$;

(b) exponential distribution $\mathscr{E}(\lambda)$;

(c) normal distribution $\mathscr{N}(\mu, \sigma^2)$

Problem 23. Alice and Bob enter simultaneously the grocery shop. Assume that their shopping times are independent **r.v.**'s X and Y respectively with exponential distribution of mean value 15 minutes.

(a) What is the distribution of the **r.v.** $M := \min(X, Y)$ for the smaller of the two shopping times? What is the expected value for the smaller shopping time?

(b) What is the distribution of the time $W := |X - Y|$ that the first to finish shopping will wait for his/her friend?

(c) Check that $\mathsf{E}(M) + \mathsf{E}(W)$ is larger than 15 min; is not there any contradiction there?

(d) Justify the strong memorylessness property: given that Alice is the second to finish her shopping, the difference $X - Y$ has the same distribution as X.

Problem 24. The joint probability density function of random variables X and Y is given by the formula

$$f_{X,Y}(x,y) = \begin{cases} a, & x,y>0, x+y<1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a

(b) Calculate the marginal **p.d.f.**'s f_X and f_Y and decide whether X and Y are independent

(c) Find $\mathsf{E}(X)$, $\mathsf{E}(Y)$ and the covariance $\mathsf{Cov}(X,Y)$

(d) Calculate $\mathsf{E}(XY)$

(e) Calculate the probability density function $f_{X|Y}(x \mid 0.5)$ of X given that $Y = 0.5$

(f) Find the conditional expectation $\mathsf{E}(X \mid Y = 0.5)$

6 New problems

Problem 1. A fair die is rolled n times. Let X_k denote the number of rolls that give k points, $k = 1, \ldots, 6$. Find

(a) the expectation $\mathsf{E}(X_k)$ and variance $\mathsf{Var}(X_k)$;

(b) the conditional distribution of X_j given that $\{X_k = m\}$ $(k \neq j)$; is it any of the standard distributions? If so, can you give an explanation for that?

(c) the covariance $\mathsf{Cov}(X_j, X_k)$; can you predict the sign of the covariance without calculations? Are the **r.v.** X_j and X_k independent?

Hint: in (c), you may condition on $\{X_k = m\}$ to calculate $\mathsf{E}(X_j X_k)$. Alternatively, calculate $\mathsf{Var}(X_j + X_k)$ and use it to determine $\mathsf{Cov}(X_j, X_k)$. Another approach can use the fact that the joint distribution of $(X_k, X_j, n - X_k - X_j)$ is multinomial $M(n; \frac{1}{6}, \frac{1}{6}, \frac{4}{6})$

Problem 2. Assume that X has a Poisson distribution $\mathscr{P}(\lambda)$ and that, conditioned on $\{X = k\}$, a **r.v.** Y is binomially distributed $\mathscr{B}(k,p)$.

(a) Find the distribution of Y;

(b) find the distribution of $X - Y$ and then prove that the **r.v.** Y and $X - Y$ are independent;

(c) find the conditional distribution of X given the event $\{Y = n\}$.

Hint: in (c), you may start with $\mathsf{P}(X = k \mid Y = n) = \mathsf{P}(X - Y = k - n \mid Y = n)$ and use part (b)

Problem 3. A number $\lambda \in (0,1)$ has binary expansion

$$\lambda = 0.b_1 b_2 b_3 \cdots = \sum_{n=1}^{\infty} b_n 2^{-n}.$$

Alice and Bob toss a fair coin repeatedly. Denote by I_n the indicator of the event that the nth toss is a Head. Let T be the first toss such that $I_n \neq b_n$, $T = \min\{n : I_n \neq b_n\}$. If $I_T < b_T$, then Alice wins; otherwise, Bob wins.

(a) Find the probability that Alice wins on the nth toss;

(b) show that the probability that Alice wins is equal to λ;

(c) find the expected number of tosses in this game.

Hint: in (c), you may derive and then use the formula $\mathsf{E}(T) = \sum_{k\geq 1} \mathsf{P}(T \geq k)$

Problem 4. Alice and Bob have a and b gold coins, respectively. They play a series of independent games in which the loser gives a gold coin to the winner; they stop when one of them has no coins remaining. Assume that Alice wins each game with probability $p \neq \frac{1}{2}$,

(a) find the probability that Alice wins;

(b) find the expected number of games played.

Hint: condition on the event W_1 that Alice wins the first game and use the total probability rule (for probabilities or expectations) to derive the *difference equation* for the winning probability p_k or average number m_k of games played when Alice starts with k coins. Find two solutions in (a); in (b), look in addition for a partial solution for m_k in the form $m_k = ck$; then use the boundary conditions to find all the constants.

Problem 5. (a) A fair coin is tossed repeatedly. Let T be the number of tosses required until three consecutive Heads appear for the first time. Find $\mathsf{P}\{T = n\}$ and $\mathsf{E}(T)$.

(b) A biased coin (showing Heads with probability p) is tossed repeatedly. Let X be the number of tosses required for the first occurrence of HHH. Find the probability generating function $\mathsf{E}(s^X)$ and then show that

$$\mathsf{E}(X) = \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3}.$$

Hint: partition the sample space into $A_1 =$"first toss T", $A_2 =$"first two tosses HT", $A_3 =$"first three tosses HHT" and the complement and use the total probability rule for expectations

Problem 6. Alice flips a biased coin (with probability p of showing Heads) 5 times, and if Tail shows up on the last toss, then she continues tossing until the first Head appears. Let N be the total number of tosses made and X be the total number of Heads.

(a) Find the distributions of **r.v.**'s N and X;

(b) find the expected values of N and X;

(c) what is the covariance of N and X? are these **r.v.**'s independent?

(d) find the conditional distribution of N given $X = 3$ and the expected value $\mathsf{E}(N \mid X = 3)$.

Problem 7. Consider a game, in which a fair coin is flipped, and if it shows Heads, a standard normal random variable $X \sim \mathcal{N}(0, 1)$ is sampled; otherwise, $X \sim \mathcal{N}(10, 1)$ is used. The value of X is reported to the player, who should then guess the face of the coin, Heads or Tails. The player decides for the following strategy: if $X \leq c$, she reports Heads, otherwise she guesses Tails.

(a) Determine the probability to guess correctly. For what c is that probability maximal?

(b) Assume that the coin is biased and shows Heads with probability p. Determine the probability to guess correctly. For what value of c is that probability maximal?

(c) In part (b), find the value of c so that the probability of a wrong guess of Heads equals the probability of a wrong guess of Tails.

Problem 8. The field hospital is located at the centre of a circle of radius 100 km. If an accident occurs within this circle, then the hospital sends out an ambulance helicopter. Assuming that an accident occurs at a point that is uniformly distributed in the circle,

(a) find the distribution of the travel distance D to the accident point, its expected value and variance;

(b) find the distribution of the abscissa X of the accident point (X, Y), its expected value and variance.

Hint: in (a), start with finding the c.d.f. of D

Problem 9. Suppose stop lights at an intersection alternately show green for one minute, red for one minute (ignore yellow). Suppose a car arrives at the lights at a time distributed uniformly at random relative to this cycle. Let X be the delay of the car at the lights, neglecting any delay due to traffic congestion.

(a) Find a formula for the **c.d.f.** of X and sketch its graph.

(b) Is X discrete, continuous, or neither?

(c) Find $\mathsf{E}(X)$ and $\mathsf{Var}(X)$.

Problem 10. Suppose that two points A and B are independently chosen at random on the unit interval. We identify the points and their (random) abscissa, so that A and B are random variables with distribution $\mathscr{U}(0, 1)$. Find the distributions of the following random variables:

(a) the distance $D := |A - B|$ between these points;

(b) $M := \max\{A, B\}$;

(c) $N := \min\{A, B\}$;

(d) which among the random variables A, B, D, M, N are pairwise independent?

(e) reconsider part (a) for the case when the points A and B are chosen at random on a unit circle.

Problem 11. Assume that X_1 and X_2 are independent **r.v.**'s with exponential distribution $\mathscr{E}(\lambda)$. Set $Y_1 = X_1 - X_2$ and $Y_2 = X_2$. Find the joint distribution of the **r.v.**'s Y_1 and Y_2 and marginal distribution of Y_1.

Hint: use formula (7.1), Section 6.7 of Ross. Y_1 has a *double exponential*, or *Laplace* distribution

Problem 12. Assume that X and Y are non-negative **r.v.**'s with a joint distribution given by the density

$$f_{X,Y}(x,y) = \begin{cases} ce^{-\lambda x}, & \text{if } 0 \le y \le x < \infty \\ 0 & \text{otherwise} \end{cases}$$

for some $\lambda > 0$.

(a) Find the constant c;

(b) determine the marginal distributions of X and Y and identify them;
(Hint: one of them is a Gamma distribution)

(c) calculate $\mathsf{E}(X)$, $\mathsf{E}(Y)$, $\mathsf{E}(X^2)$, $\mathsf{E}(Y^2)$;

(d) calculate $\mathsf{Var}(X)$, $\mathsf{Var}(Y)$, $\mathsf{Cov}(X,Y)$;

(e) determine the conditional distribution of Y given that $X = x$ and identify it.

(Hint: using the gamma-functions eliminates almost all calculations)